Cheatsheet examen practic USO

CONFIGURATIE SSH PENTRU FEP

```
ssh-keygen -t rsa
ssh-keyscan -H fep.grid.pub.ro >> ~/.ssh/known_hosts
ssh-copy-id -i ~/.ssh/id_rsa.pub user@fep.grid.pub.ro
touch ~/.ssh/config
Host fep
   HostName fep.grid.pub.ro
   User nume_user
```

UTILIZATORI SI GRUPURI

SCHIMBARE HOME PENTRU USER
```
usermod -d /newhome/username username
```

SCHIMBARE SHELL
```
sudo usermod --shell /bin/bash user
sudo userad --shell /bin/bash user
```

ADAUGARE USER LA GRUP
```
sudo useradd -a -G grup user
gpasswd -a user grup
```

STERGERE USER DE LA GRUP
```
gpasswd -d user grup
```

SCHIMBARE PAROLA USER
```
passwd user
```

SETEAZA PERMISIUNI UTILIZATOR
```
setfacl -m u:john:r-x file1.txt
```

TAR.GZ

COMPRIMARE TAR.GZ
```
tar -cvzf arhiva.tar.gz fisier1 fisier2
```

DECOMPRIZARE TAR.GZ
```
tar -xvf arhiva.tar.gz
```

FORMATARE SI MONTARE

FORMATARE
```
mkfs mkfs.bfs mkfs.cramfs mkfs.ext2 mkfs.ext3 mkfs.ext4
mkfs.fat mkfs.minix mkfs.msdos mkfs.ntfs mkfs.ufat
```

MONTARE
```
sudo mount fisier_montat director
--read-only
--read-write
```

DEMONTARE
```
sudo umount director
```

SERVICES, PROCESSES, DEVICES, HARDWARE

TOATE SERVICIILE PORNITE (RUNNING)
```
systemctl --type=service --state=running
```

States posibile:
```
dead, activating, activating-done, active, deactivating, deactivating-sigterm,
deactivating-sigkill, failed, cleaning
dead, active
dead, waiting, running, elapsed, failed
```

AFISARE DEVICE DE TIP BLOCK
```
lsblk -dn
```
(fara dn afiseaza si header de informatii si toate sub-devices)
```
ls -l /dev | grep '^b' | awk '{print $10}'
```

AFISARE DEVICE DE TIP CHARACTER
```
ls -l /dev | grep '^c' | awk '{print $10}'
```

AFISARE INFORMATII DESPRE PROCESOR
```
lscpu
cat /proc/cpuinfo
```

SUMA PID-URILOR
```
ps -eo pid | awk '{s+=$1} END {print s}'
```

PID-UL PROCESULUI CARE CONSUMA CEA MAI MULTA MEMORIE
```
ps -e --no-header -o pid --sort -%cpu | head -1
```

IDENTIFICATORI PS
```
pcpu, group, ppid, user, args, comm, rgroup, nice, pid, pgid, etime, ruser, time,
tti, vsz
%cpu, %mem
```

SECURITATEA SISTEMULUI

CRIPTARE CU OPENSSL
```
openssl aes-256-cbc -in plaintext_file.txt -out encrypted_file.enc -pass
pass:"uso rules"
```
(pentru decriptare se adauga -d)

GENERARE PAROLE
```
pwgen
```

SCRIPTING

CITIRE FISIER LINIE CU LINIE
```
while read line; do
  echo $line
done < file.txt
```
dacă are un anumit separator se foloseste IFS=, (sau orice alt caracter)

TEST COMMAND

LUCRUL CU LINII

REPLACE CHARS
```
sed 's/old_string/new_string/' filename.txt
```

REMOVE LAST CHAR
```
echo 'Remove this question mark?' | rev | cut -c 2- | rev
```

INTOARCE LINIILE
```
nl /etc/passwd | sort -nr | cut -f 2-
```

LINII CARE NU CONTIN UN CUVANT
```
grep -v 'cuvant'
```

ALTE UTILE

NUMELE TUTUROR VARIABILELOR DE MEDIU
```
printenv | cut -d '=' -f 1
```

GENERARE STRING RANDOM
```
tr -cd 'A-Za-z0-9' < /dev/urandom | head -c 4
```

CAUTARE FISIERE MAI MARI DE DIMENSIUNEA ...
```
find dir -type f -size +4G
```

CONECTARE NEINTERACTIVA
```
sudo -u user bash -c 'echo "I am $USER with UID $UID"'
```

AFISARE FISIERE DIN DIRECTORUL CURENT (FARA DIR)
```
find . -maxdepth 1 -type f
```

AFISEAZA HH:MM:SS ULTIMULUI COMMIT REALIZAT
```
git log -1 --format=%cd | awk '{print $4}'
```

GENERARE NUMAR RANDOM CU SHUF
```
shuf -i 0-255 -n 1
```

IESIRE DIN SCRIPT CU EROARE
```
exit 1
```

PING CU PADDING 1
```
ping ip -p ff
```

SORT FOLOSIND DELIMITATOR SI COLOANA
```
sort -t ':' -k 1,1
```

AFISARE INFORMATII FISIER / DIRECTOR
```
stat nume
```

GENERARE FISIER DE O ANUMITA MARIME
```
dd if=/dev/zero of=path bs=100M count=1
```

FISIER DE O ANUMITA MARIME CU ANUMIT CARACTER
```
dd if=/dev/zero bs=1M count=100 | tr "\000" "\006" > file.txt
```